

Lab Assignment 6: Transfer Function Analysis

Revision October 01, 2018

Summary

In this lab, we employ transfer functions to determine the frequency response and step response of second-order passive and active circuits, including a second order RLC high pass filter, a second order Butterworth low pass filter, and a wide-band band pass filter.

Learning Outcomes: After completing this laboratory assignment, you should be able to:

- Obtain the transfer function of a passive RLC second order high pass filter, and use it to determine the filter's amplitude and phase responses as a function of frequency, as well as the filter's step response.
- Design a passive second order high pass filter to meet design specifications.
- Use the transfer function of an active second order filter to design a Butterworth low pass filter to meet design specifications, and to obtain the filter's amplitude, phase and step responses.
- Use the transfer function to find the amplitude, phase and step responses of the wideband band pass filter formed by serially concatenating the passive RLC second order high pass filter with the active second order Butterworth low pass filter.
- Compare laboratory measurements of the amplitude, phase and step responses of the above described circuits with the theoretical responses predicted by their transfer functions.

Required Equipment

- EE352 Analog parts kit
- Breadboard
- Function Generator
- Oscilloscope
- DC Power Supply

I. Second Order RLC High Pass Filter

In this part of the lab assignment, we use the transfer function to obtain a second order RLC high pass filter's amplitude and phase responses as a function of frequency, as well as the filter's step response. We then design and build the circuit (shown in Fig. 1) and verify that the experimental responses match those predicted by the circuit's transfer function.

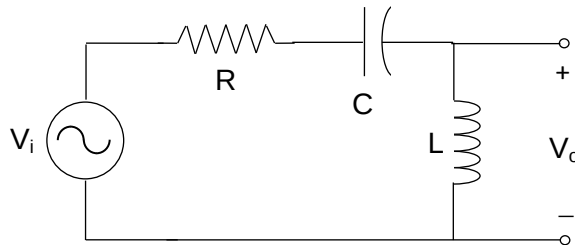


Figure 1. High-pass RLC circuit.

Pre-Lab:

- For the high-pass RLC circuit shown in Fig. 1, use your knowledge of the frequency-dependent behavior of circuit components to predict the output voltage V_o when the input frequency $\omega = 0$.
- Repeat part (a), except now let $\omega \rightarrow \infty$.
- Based on your answers to parts a and b, what type of filter is this?
- By analyzing the circuit in the s-domain, derive the transfer function $H(s)$ for this circuit.
- The denominator of the transfer function for a 2nd order circuit is $D(s) = s^2 + 2\zeta\omega_n s + \omega_n^2$, where ω_n is the natural frequency, and ζ is the damping factor. The circuit is overdamped when $\zeta > 1$, critically damped when $\zeta = 1$, and underdamped when $\zeta < 1$. Using these facts, choose component values R , L and C for the 2nd order circuit in Fig. 1 such that the natural frequency $f_n = 20\text{kHz}$, and the damping factor $\zeta = 0.2$.
- Find the frequency response $H(j\omega)$ of the circuit, and the amplitude response $|H(j\omega)|$. What are the values of the frequency response $H(j\omega)$ when $\omega = 0$ and when $\omega \rightarrow \infty$?
- What is the slope of graph of amplitude response $|H(j\omega)|$ in dB vs. $\log_{10}(\omega)$ at low ($\omega \ll \omega_n$) and high ($\omega \gg \omega_n$) frequencies? Sketch the graph of amplitude response $|H(j\omega)|$ in dB vs. $\log_{10}(\omega)$.
- Analyze the behavior of the phase response $\angle[H(j\omega)]$ at low ($\omega \ll \omega_n$), high ($\omega \gg \omega_n$) and intermediate ($\omega = \omega_n$) frequencies. Sketch the graph of the phase response $\angle[H(j\omega)]$ vs. $\log_{10}(\omega)$.
- Use either MATLAB or LTSPICE to graph the amplitude, phase, and step responses for this circuit, for the component values you chose in part e. Include these graphs in your lab notebook, as well as the MATLAB code or LTSPICE circuit you used to produce the graphs.

Lab Procedures:

- Construct the circuit in Fig. 1, using the component values you chose in part (e) of the pre-lab. Measure and record the actual values of L and C . *Important note: You should leave*

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out the 50 Ω resistor R when you build the circuit in Fig. 1. The signal generator's internal 50 Ω source resistance will serve as the resistor R .

- Experimentally measure the frequency response of your circuit (magnitude and phase) by using a sine wave of amplitude 8V p-p as the input to the circuit. Measure the amplitude and phase responses for at least 10 different frequencies, from a frequency range of at least an order of magnitude below the natural frequency to an order of magnitude above the natural frequency. Plot your results in your lab notebook, on a semi-log scale (i.e., with $\log_{10}$ of frequency on the horizontal axis and gain or phase on the vertical axis) with units of gain as dB and units of phase as degrees. Note: You can measure phase responses as $\text{angle}[H(j\omega)] = \text{angle}[V_o(j\omega)/V_i(j\omega)] = \text{angle}[V_o(j\omega)] - \text{angle}[V_i(j\omega)]$ by measuring the time delay between the output and input signals. For example, if the peak in the output sinusoid leads (trails) the peak in the input sinusoid by Δt seconds, then

$$\text{angle}[H(j\omega)] = + (-) 360 \times \Delta t/T = + (-) 360 \times f\Delta t,$$

where $T = 1/f$ is the period of the input sine wave, and the angle units are degrees.

- Experimentally determine the step response of your circuit by inputting a 1V p-p square wave with a DC offset of 0.5V. The frequency should be set high enough to show the step response details, but low enough so that the final value of the step response is clear. Recommend starting at a frequency of $1/10^{\text{th}}$ the natural frequency. Sketch the step response in your lab notebook and save the data to a flash drive for later processing. Determine the period of any oscillations in the response and compare it to the natural frequency of the circuit.
- Demonstrate operation of your circuit to the Teaching Assistant for both the sinusoidal input and step input cases. Have the TA initial your lab checklist.*
- After completing Part I, leave the RLC circuit on your breadboard, as it will be used again in Part III of this lab assignment.
- In your lab report, make sure to graph your experimental amplitude, phase and step responses and compare them with the graphs you found in step (i) of the pre-lab.

II. Second Order Active Butterworth Low Pass Filter

In this part of the lab, you will design and build a second order active Butterworth low pass filter, based on the 2nd order op-amp low pass filter circuit shown in Fig. 2. Here are a few facts about Butterworth low pass filters. For more details please see the filter design background reading assignment posted at the EE352 course website.

- An Nth order Butterworth lowpass filter has magnitude squared frequency response

$$|H(j\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2N}}$$

where ω_c is the 3-dB cutoff frequency.

- The transfer function $H(s)$ is characterized in terms of $H(s)H(-s)$ as

$$H(s)H(-s) = \frac{1}{1 + (-s^2/\omega_c^2)^N}.$$

- The N poles s_k of the transfer function (i.e., the values of s that make the denominator zero) occur on a circle of radius ω_c at equal angular spacing:

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$$s_k = \omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N}, \quad k = 0, 1, \dots, N-1.$$

Complex valued poles of $H(s)$ always occur in complex conjugate pairs.

- The amplitude response of a Butterworth lowpass filter is flat at low frequencies (i.e., in the filter's passband), and then decreases monotonically until falling below the stopband attenuation δ_2 at the desired stopband frequency ω_s .
- For a given cutoff frequency ω_c , stopband frequency ω_s , and stopband attenuation δ_2 , the filter order N is determined as

$$N = \left\lceil \frac{\ln[(1/\delta_2^2) - 1]}{2 \ln(\omega_s/\omega_c)} \right\rceil,$$

where $\lceil x \rceil$ means round x up to the closest integer $\geq x$.

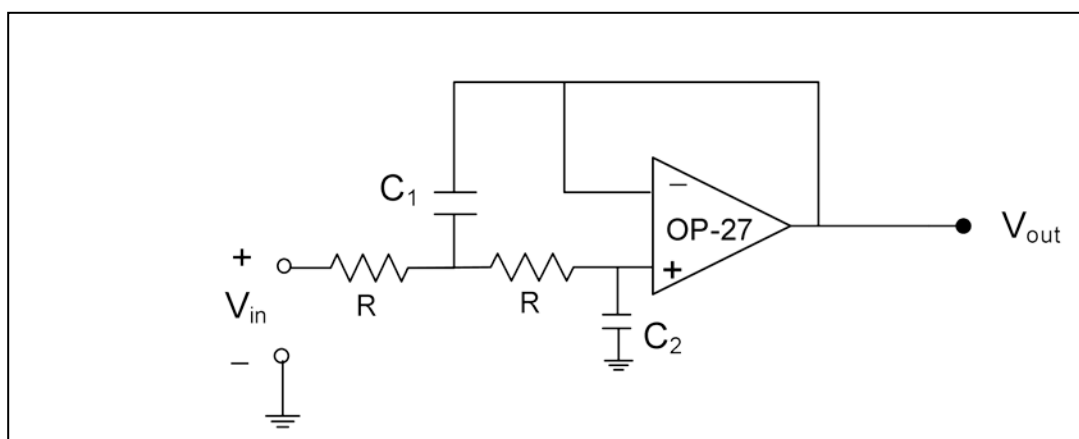


Figure 2. 2nd order op-amp low pass filter.

Figure 2 shows a second order active low pass filter. It can be shown that the transfer function of this filter is

$$H(s) = \frac{\frac{1}{R^2 C_1 C_2}}{s^2 + \frac{2}{RC_1}s + \frac{1}{R^2 C_1 C_2}} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2},$$

where ω_n is the natural frequency and ζ is the damping factor.

Pre-lab:

- Design a Butterworth low pass filter with cutoff frequency $f_c = 80$ kHz, stopband frequency $f_s = 300$ kHz, and stopband attenuation of -20 dB. Your design should specify the filter order and the filter pole locations.
- After verifying that the order N of your Butterworth filter is $N = 2$, select component values R , C_1 and C_2 so that your Butterworth filter can be implemented with the circuit in Figure 2.
- Using MATLAB or PSPICE, plot the amplitude, phase, and step responses of your Butterworth lowpass filter. Include MATLAB code or LTSPICE circuit setup.

Lab Procedures:

1. Construct the circuit shown in Figure 2 using the component values you selected in part (b) of the pre-lab. Make sure that the OP27 is connected to +12V and -12V power supplies and its ground is connected to the ground of the rest of the circuit. Measure and record the values of R , C_1 and C_2 in your lab notebook. Since your RLC circuit from Part I should still be on your breadboard, make sure that your Butterworth filter circuit is constructed on a different section of your breadboard than the circuit of Part I.
2. Experimentally measure the frequency response of your circuit (magnitude and phase) by inputting a sine wave of amplitude 8V p-p. Measure the amplitude and phase responses for at least 10 different frequencies, from a frequency range of at least an order of magnitude below the natural frequency to an order of magnitude above the natural frequency. Plot your results in your lab notebook, on a semi-log scale with units of gain as dB and units of phase as degrees.
3. Experimentally determine the step response of your circuit by inputting a 1V p-p square wave with a DC offset of 0.5V. Sketch the step response in your lab notebook and save the data to a flash drive for later processing. Determine the period of any oscillations in the response and compare it to the natural frequency of the circuit.
4. *Demonstrate operation of your circuit to the Teaching Assistant for both the sinusoidal input and step input cases. Have the TA initial your lab checklist.*
5. In your lab report, make sure to graph your experimental amplitude, phase and step responses and compare them with the graphs you found in step (c) of the pre-lab.
6. At the end of Part II, the circuits from both Part I and Part II should both be on your breadboard.

III. Wide-band Band Pass Filter

In this part of the lab, you will serially connect the high pass filter from Part I with the low pass filter from Part II, in order to construct a wide-band bandpass filter with a lower cutoff frequency of approximately 20 kHz, and an upper cutoff frequency of approximately 80 kHz. If we denote the transfer function of your highpass filter from Part I as $H_h(s)$, and the transfer function of your lowpass filter from Part II as $H_l(s)$, then the transfer function $H_b(s)$ of the fourth-order bandpass filter you will construct will be $H_b(s) = H_h(s) H_l(s)$.

Pre-lab:

- (a) Compute the transfer function $H_b(s)$ of your wide-band bandpass filter. (Note: This can easily be done using MATLAB's conv command.)
- (b) Use MATLAB to plot the theoretical amplitude, phase, and step responses of your wide-band bandpass filter. Include MATLAB code in your pre-lab solution.

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Lab Procedures:

1. Hook the output of your RLC highpass filter from Part I to the input of a voltage follower circuit as shown in Figure 3 below. The voltage follower should be constructed with the OP-27 op-amp. The voltage follower circuit can be found in the last section of Module 3.3 used in Lab3, which is available on the course website. All op-amps should use +12V and -12V power.

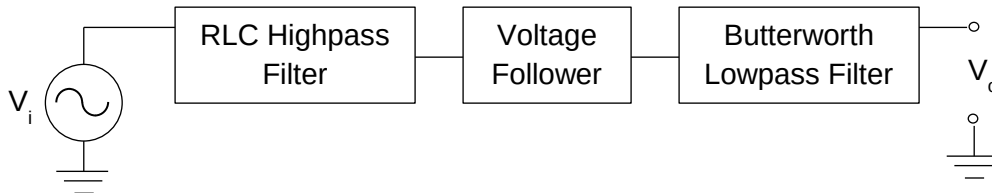


Figure 3. Wide-band bandpass filter.

2. Hook the output of the voltage follower to the input of your Butterworth lowpass filter circuit from Part II. The voltage follower ensures that the high pass filter sees a very high load resistance, and that the lowpass filter sees a very low source resistance.
3. Experimentally measure the amplitude frequency response (but not the phase response) of your bandpass filter circuit by inputting a sine wave of amplitude 4V p-p to the RLC highpass filter, and measuring the output of your Butterworth lowpass filter. Measure the amplitude response for at least 20 different frequencies, from a frequency range of at least an order of magnitude below the highpass filter's cutoff frequency to an order of magnitude above the lowpass filter's cutoff frequency. Plot your results in your lab notebook, on a semi-log scale with units of gain as dB.
4. Experimentally determine the step response of your circuit by inputting a 1V p-p square wave with a DC offset of 0.5V. Sketch the step response in your lab notebook and save the data to a flash drive for later processing. Determine the period of any oscillations in the response and compare it to the natural frequencies of both the high pass and lowpass filter circuits.
5. *Demonstrate operation of your circuit to the Teaching Assistant for both the sinusoidal input and step input cases. Have the TA initial your lab checklist.*
6. In your lab report, make sure to graph your experimental amplitude and step responses and compare them with the graphs you found in step (b) of the pre-lab.

Lab Report:

In your lab report, provide a summary of the results of this lab assignment. You should include, at a minimum, all items indicated on the lab checklist. Append the lab checklist sheet with Laboratory Instructor's initials indicating completed lab demos to your report.

Lab 6 Checklist:

Name: _____

Lab title and introduction

(5 pts)

- Lab title, your name, date, and lab partner.
- Brief introduction (two or three sentences) explaining the purpose of this lab.

I. RLC Highpass Filter (40 pts total)

1. Diagram of circuit with values of R, L and C to obtain desired damping ratio and natural frequency. (3 pts)
2. Circuit transfer function as a ratio of polynomials in s with numerical coefficients. (3 pts)
3. Transfer function behavior at high and low frequencies. (2 pts)
4. Measured R, R_L , L, and C for circuit implementation. (5 pts)
5. Measured amplitude and phase responses (in graphical form). (10 pts)
6. Measured step response (presented in graphical form). (4 pts)
7. Comparison between measured frequency of oscillations in step response and natural frequency (include % difference, assessment of linearity of circuit) (3 pts)
8. Simulated frequency and step responses (MATLAB or PSPICE). (5 pts)
9. **DEMO:** Have a TA initial this sheet, indicating that he/she has observed your circuit's operation. (5 pts)

II. Second Order Active Butterworth Low Pass Filter (40 pts total)

1. Butterworth filter circuit diagram. (3 pts)
2. Designed circuit transfer function with numerical coefficients. (4 pts)
3. Theoretical resistance, capacitance values for your design. (4 pts)
4. Measured resistance, capacitance values (2 pts)
5. Measured amplitude and phase responses in graphical form. (10 pts)
6. Measured step response in graphical form. (4 pts)
7. Simulated frequency and step responses (MATLAB or PSPICE). (4 pts)
8. Comparison of theoretical and measured responses (4 pts)
 - Cutoff frequency
 - Passband (low frequency) gain
 - Slope of magnitude response above cutoff frequency
 - Step response
9. **DEMO:** Have a TA initial this sheet, indicating that he/she has observed your circuit's operation. (5 pts)

III. Wide-band Bandpass Filter (15 pts total)

1. Block diagram of circuit (similar to Fig. 3). (1 pts)
2. Measured amplitude response in graphical form. (4 pts)
3. Measured step response in graphical form. (2 pts)
4. Simulated (MATLAB) amplitude and step responses. (2 pts)
5. Compare measured and simulated responses: (3 pts)
 - Cutoff frequencies (low and high)
 - Step response oscillation frequency compared to both filters' natural frequencies.
6. **DEMO:** Obtain TA signature indicating the circuit works. (3 pts)